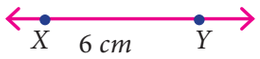
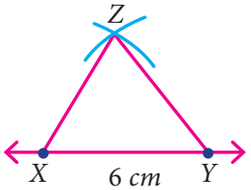
Step 2: With X as centre, draw an arc of radius 5 *cm* above the line XY.



Step 3: With Y as centre, draw an arc of radius 5.5 *cm* to intersect arc drawn in step 2. Mark the point of intersection as Z.

Step 4: Join XZ and YZ.

Now XYZ is the required triangle.



Using trace copy superposition method, if we place the $\triangle XYZ$ on the $\triangle PQR$ (Fig. 4.24) in such a way that the sides PQ on XY, PR on XZ and QR on YZ, then both the triangles ($\triangle XYZ$ and $\triangle PQR$) matches exactly in size and shape.

Here we used only sides to check the congruence of the triangles, so we can say the condition as, if three sides of one triangle are equal to the corresponding sides of the other triangle then the two triangles are congruent. This criterion of congruency is known as Side – Side – Side.

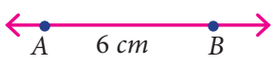
Note

It two triangles are congruent then their corresponding parts are congruent. We can say "Corresponding Parts of Congruent Triangles are Congruent". It can be abbreviated as CPCTC.

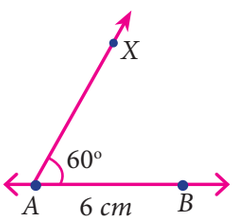
2. Side-Angle-Side congruence criterion (SAS) - The lengths of two sides and the angle included between the two sides are given.

Draw a triangle ABC given that $AB = 6\ cm$, $AC = 5\ cm$ and $\angle A = 60°$

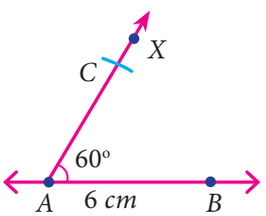
Step 1: Draw a line. Mark A and B on the line such that $AB = 6\ cm$



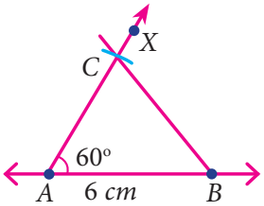
Step 2: At A, draw a ray AX making an angle of 60° with AB.



Step 3: With A as centre, draw an arc of radius 5 *cm* to cut the ray AX. Mark the point of intersection as C.



Step 4: Join BC.



7th_Maths_T2_EM_Chp4.indd 80 29-05-2019 12:19:40